TEAM 11 / MAIE 6000C

File Trace

Read-only local retrieval with source evidence

W7 midterm milestone / integration branch / 17 October 2026

Users remember meaning, not filenames

W7 proves the retrieval path before adding wider formats or generation.

A local file assistant should answer a simple question: where did this idea appear?

The result is a file, a version, a page and a passage. The original file stays in place.

Authorized root
Read-only mount

Text PDFs
Page-located chunks

Keyword search
5/5 fixed queries

Visible failure
No silent omission

Out of scope: OCR, automatic monitoring, file rewrite, graph and generated answers.

One vertical path from folder to evidence

1	**Root**	Choose a configured read-only mount
2	**Job**	Return root_id and job_id before work starts
3	**Scan**	Hash files and publish current versions
4	**Parse**	Store page-located chunks
5	**Search**	Return the current chunk and open its page

The API creates durable intent. The worker owns processing. The client reads committed state.

Version identity makes repeat scans safe

16
files in the frozen corpus

0
new files on repeat scan

0
original hash changes

The scanner records a stable file row and an immutable content version. A repeated scan reuses the same version and checks for an active extraction job before creating another one.

Live root: 12 basic-ready, 0 failed, 4 abnormal or unsupported records with reason codes.

Page numbers are checked, not assumed

12/12

sampled passages found on their claimed page

20/20

fragments found on their claimed page

0

corrupt characters in the quality sweep

A chunk never crosses a page boundary. Encrypted, damaged, image-only, missing and changing files become distinct outcomes, not one generic failure.

Page attribution used a different PDF library from the production extractor.

Human decisions survive reprocessing

Candidate

Person, project or topic
with page, quote and offsets

Decision

Confirm, correct or reject
with a revision guard

History

Revision 1 remains stored
after revision 2 exists

A stale reviewer receives a conflict instead of overwriting another person's decision.

The relation layer is a bounded candidate list, not a knowledge graph.

Search reads one canonical chunk model

source_chunks
B's canonical text

1

chunk_search_index
terms + optional vector

2

POST /search
current authorized hits

The index row is keyed by source_chunks.id and cascades when the chunk is replaced. Text remains the source of truth. The vector is optional derived data.

The search response carries file, version, chunk, page and text so evidence can reopen the exact passage.

Keyword and semantic search are both live.

Keyword

5/5

frozen queries hit the expected file and page

Semantic

5/5

20 chunks embedded
frozen descriptions hit file and page

If the model is missing, the API returns keyword results with an explicit degradation reason. It never calls that a semantic success.

Semantic mode is verified with the prepared model; fallback remains explicit when it is unavailable.

The browser calls the real services

01	**Choose a read-only root**	The client sends a relative path inside the configured mount.
02	**Watch committed job state**	The screen shows processed, pending, searchable, failed and unsupported.
03	**Search the current versions**	The result includes a match reason and a page locator.
04	**Open version-bound evidence**	The inspector shows the file, version id, page and stored passage.

The acceptance run used the real API, PostgreSQL, worker and parser. Mock screenshots are not counted.

M1 evidence is visible in the product

16 / 16	12	4	5 / 5
files processed	searchable files	reasoned attention items	keyword file and page

Opened evidence: team11-demo-01.pdf / version f3ec6956... / page P1 / stored passage

Repeat scan: 0 new files or versions. Original corpus SHA-256 changes: 0.

What is not claimed yet

Semantic quality

Pinned E5 model prepared. Twenty chunks embedded and S01-S05 hit 5/5 with matching evidence.

Worker recovery

Retry, backoff and leases are implemented. Competing-worker load testing remains open.

Release package

Deck, recording, final README, QA and manifest must match one frozen commit.

The report records a P0/P1 boundary: no disabled semantic path may masquerade as a successful result.

A narrow path with a source for every result

The folder is authorized. The version is durable.
The page is checked. The failure is visible.

Questions and scope discussion